AWS Level 2 terminal documentation

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I completed and verified an existing EC2 server using AWS CLI, SSH, Nginx and Git. The website returned HTTP 200 and displayed Hey me in browser and terminal tests. After testing, the instance was terminated as instructed in the guide.

CloudTrail records that the instance was created through AWS CLI on 22 September 2026 at 16:02:31 Swedish time. Configuration, deployment and tests for this submission were performed on 23 October 2026.

Environment and resources

Property	Value
Region	eu-north-1
Instance name	level2-web-server
Instance ID	i-00d4bfd970ff8f5f6
Instance type	t3.micro
AMI	ami-06cfeaaa22092f09d
OS	Amazon Linux 2023 x86_64
VPC	vpc-00dcc7fb53fe9926b
Subnet	subnet-05d4dca4820d2eb94 / eu-north-1a
Security group	level2projectsecuritygroup / sg-0779a81fe1b9a62e5
Original key pair	level2-key
Public IP	13.60.73.156
Private IP	172.31.27.147
Nginx / Git	1.30.4 / 2.50.1

Reference: level2.console.complete (4)(2).sh. The guide’s resource IDs and file handling were adapted to the actual environment.

Preparation with AWS CLI

AWS CLI 2.36.50 was installed in Ubuntu under WSL. STS verified the IAM user tabish-admin, and the configured region was eu-north-1. The existing server uses the subnet in eu-north-1a.

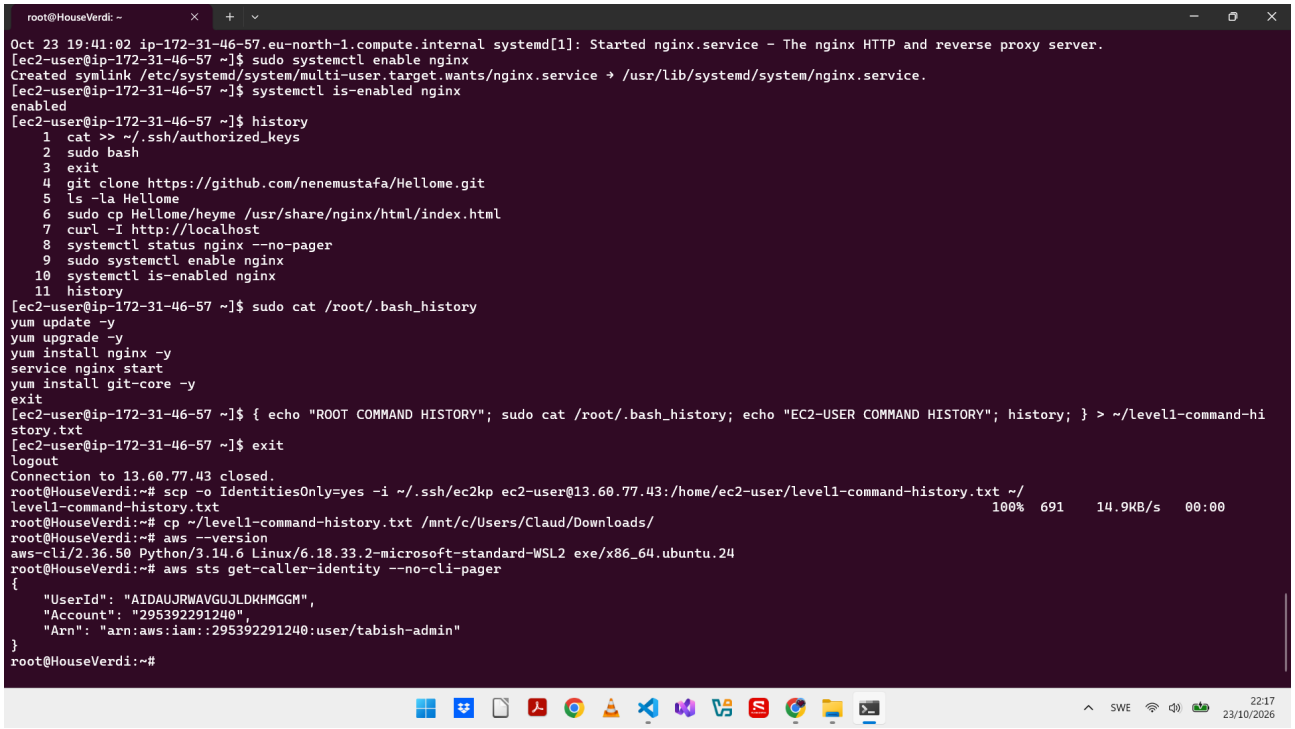


Figure 1. AWS identity verified with STS

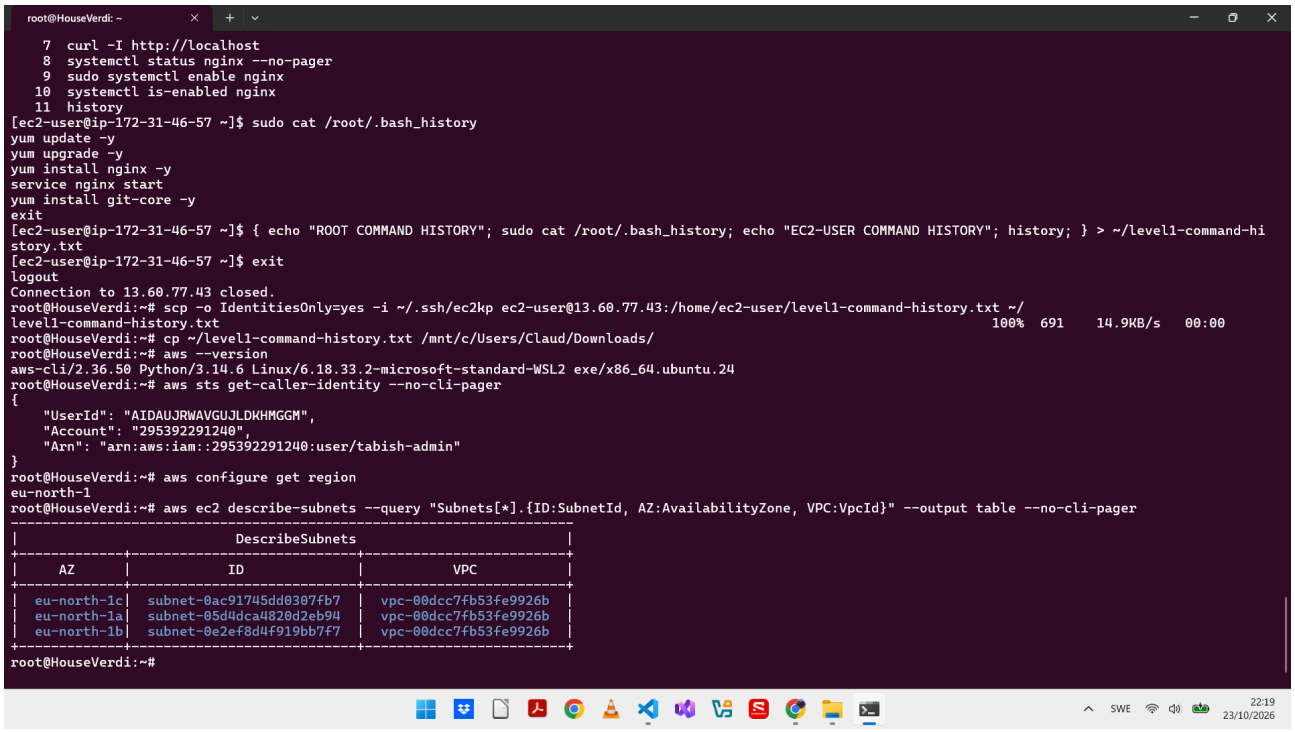


Figure 2. Available subnets and availability zones

Network and security group verification

The existing security group allowed TCP ports 22 and 80 from 83.188.29.94/32. This matched my public IP during testing. Outbound traffic was allowed to 0.0.0.0/0.

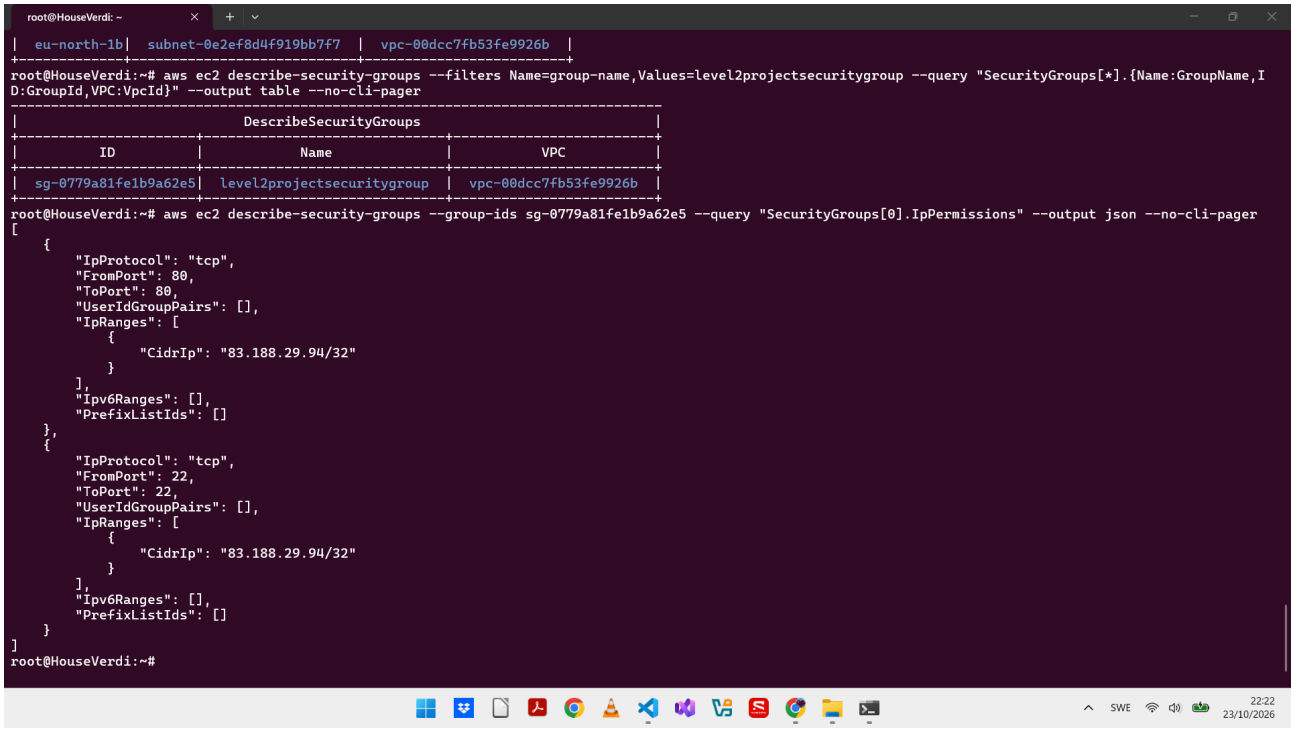


Figure 3. Inbound rules for SSH and HTTP

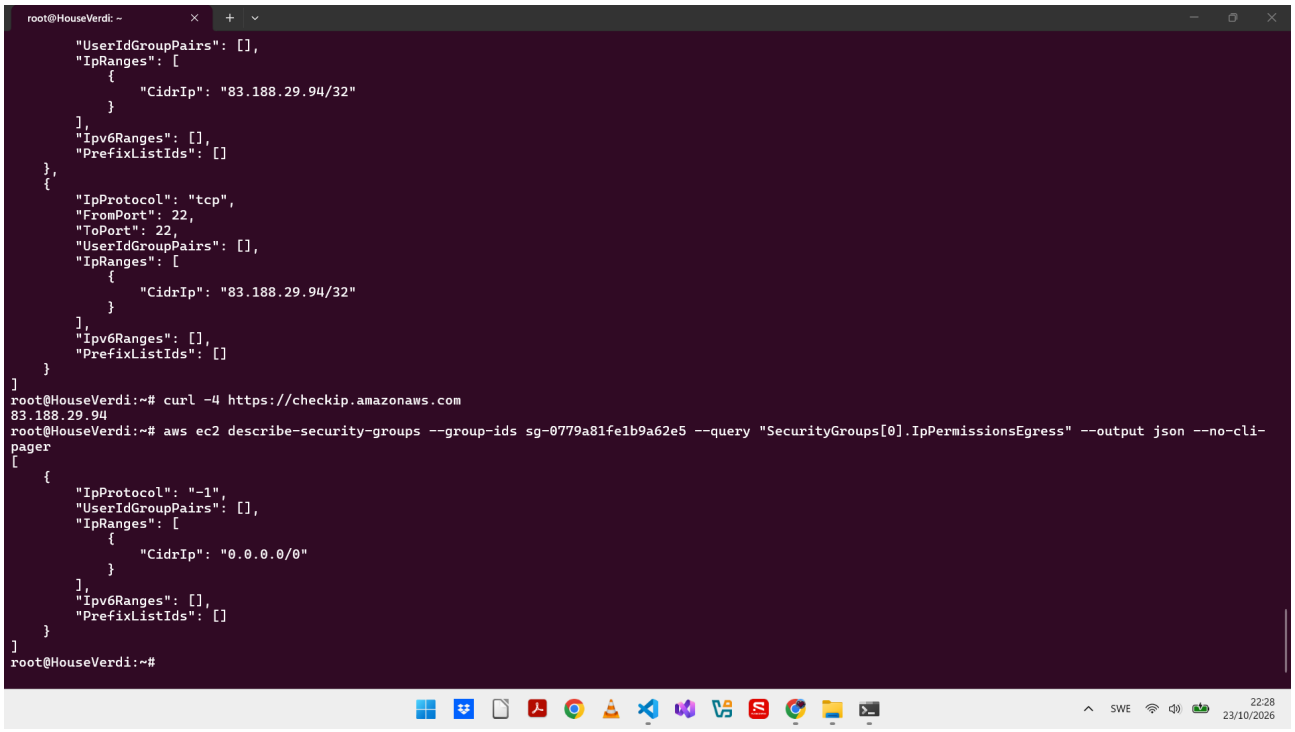


Figure 4. Outbound traffic permitted

Verification of instance creation

The CloudTrail RunInstances event records the instance ID, AMI, t3.micro type, security group, subnet and level2-key key pair. Its userAgent contains aws-cli and ec2.run-instances. This confirms CLI creation on 22 September.

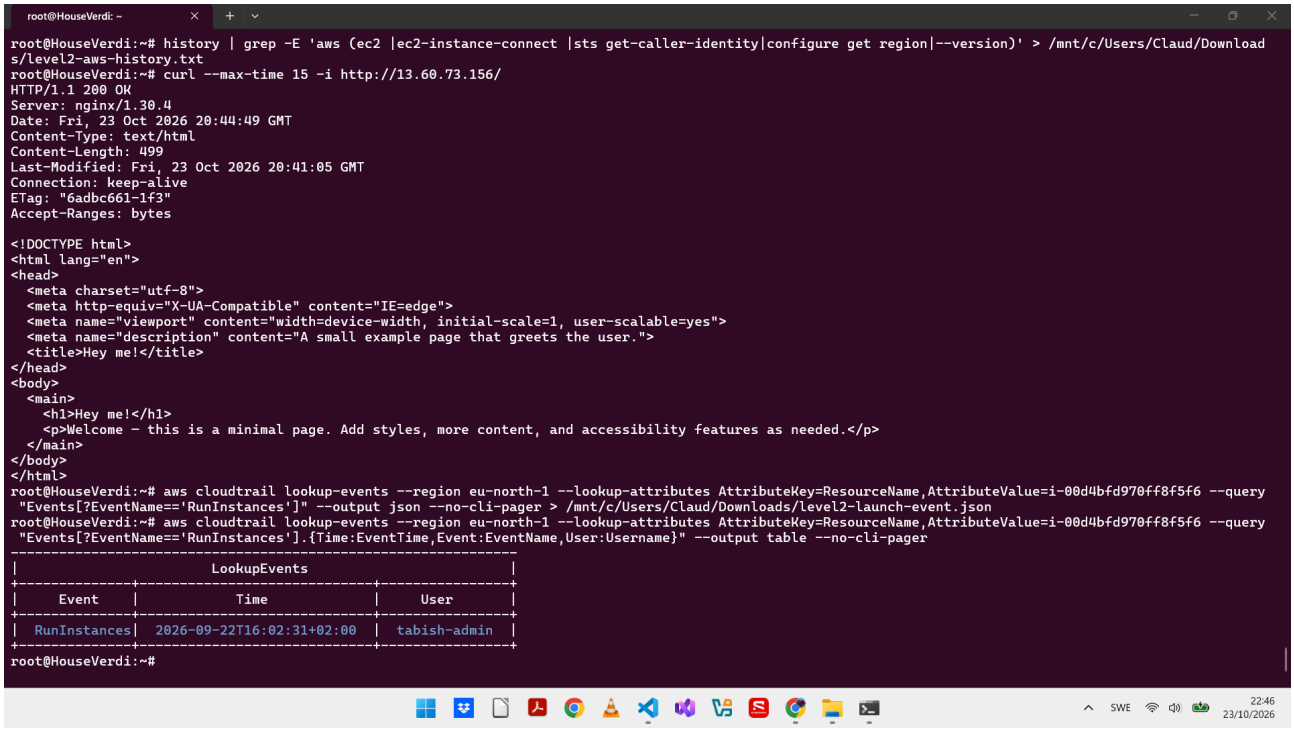


Figure 5. Creation event displayed in the terminal

The equivalent command below is reconstructed from the CloudTrail parameters. It was not rerun during this completion because the instance already existed.

```
aws ec2 run-instances --region eu-north-1 \
  --image-id ami-06cfeaaa22092f09d --instance-type t3.micro \
  --key-name level2-key --count 1 \
  --security-group-ids sg-0779a81fe1b9a62e5 \
  --subnet-id subnet-05d4dca4820d2eb94 \
  --tag-specifications \
  "ResourceType=instance,Tags=[{Key=Name,Value=level2-web-server}]"
```

SSH and restored access

The private key for level2-key was missing locally. I derived the public part of my existing local key and imported it as ec2kp-level2. This import did not change the instance's original key pair.

Access was restored by sending the public key with EC2 Instance Connect and immediately connecting through SSH as ec2-user. The private key remained local.

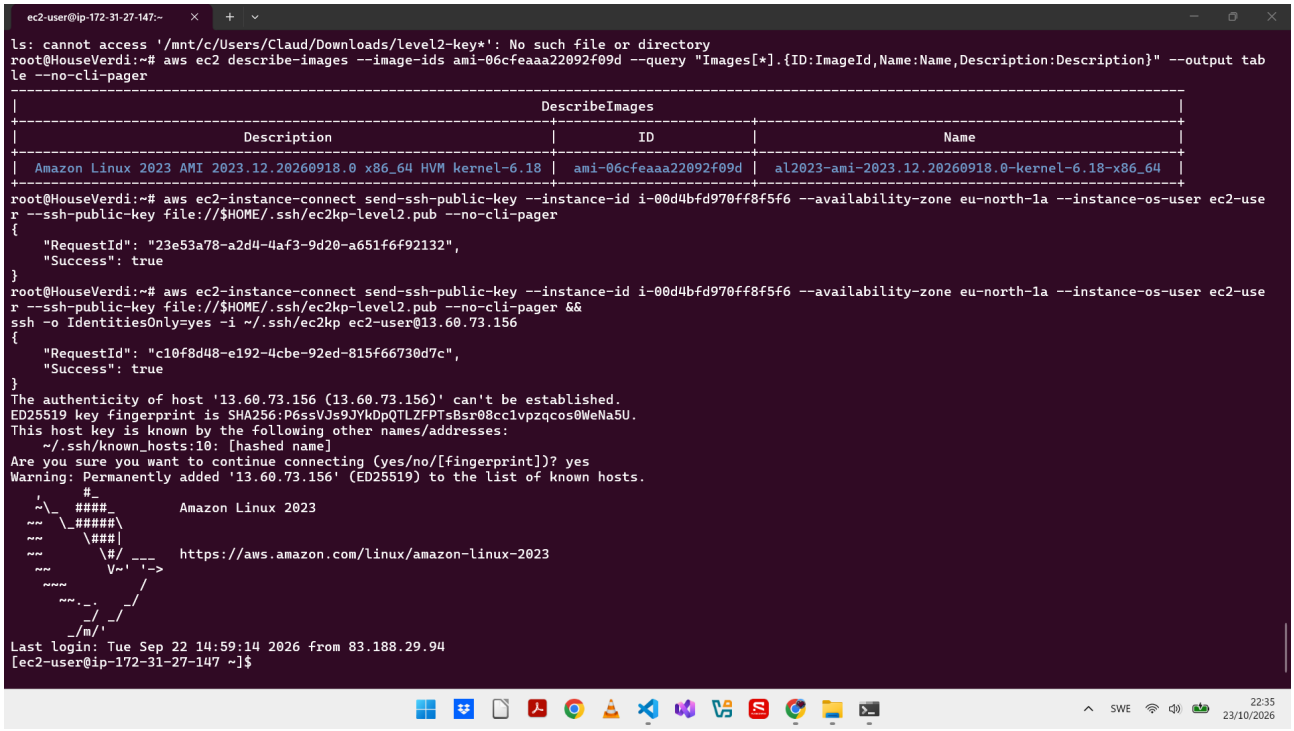


Figure 6. Successful SSH login to Amazon Linux 2023

```
ssh-keygen -y -f ~/.ssh/ec2kp > ~/.ssh/ec2kp-level2.pub
aws ec2-instance-connect send-ssh-public-key \
  --instance-id i-00d4bfd970ff8f5f6 \
  --availability-zone eu-north-1a --instance-os-user ec2-user \
  --ssh-public-key file://$HOME/.ssh/ec2kp-level2.pub \
  --no-cli-pager && \
ssh -o IdentitiesOnly=yes -i ~/.ssh/ec2kp ec2-user@13.60.73.156
```

Nginx installation and startup

The first status check returned Unit nginx.service could not be found. I installed Nginx and Git, then enabled and started Nginx. Its status became active running, and the configuration test succeeded.

```
sudo dnf install nginx git -y
sudo systemctl enable --now nginx
sudo systemctl status nginx --no-pager
sudo yum -y update && sudo yum -y upgrade
```

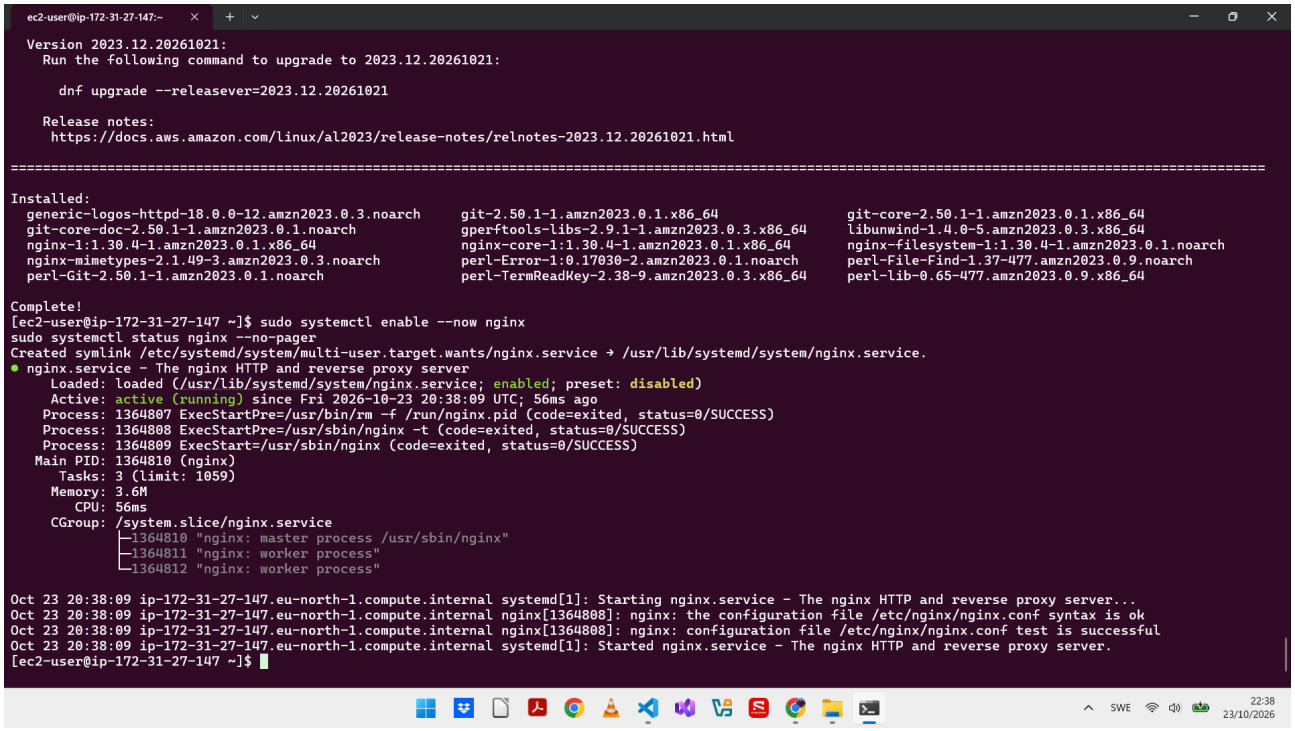


Figure 7. Installed packages and active enabled Nginx service

The update command completed with Nothing to do and Complete for the configured release. No --releasever upgrade or reboot was performed during this completion.

Website deployment

I cloned the repository and inspected its files. The website HTML was in the 499-byte file heyme. I copied it to index.html in the Nginx web root so the page is served at the root URL.

```
git clone https://github.com/nenemustafa/Hellome.git
ls -l Hellome
sudo cp Hellome/heyme /usr/share/nginx/html/index.html
curl -i http://localhost/
```

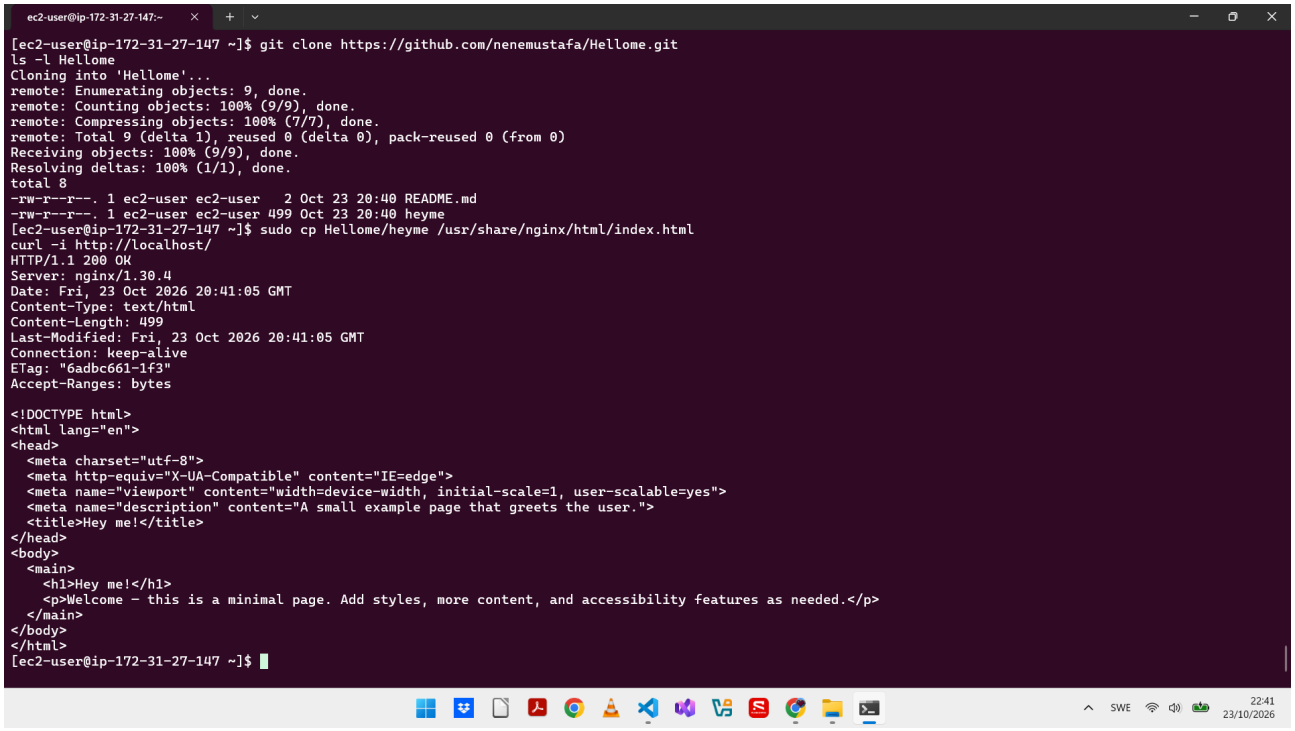


Figure 8. Deployment and local HTTP test showing the page HTML

The response was HTTP/1.1 200 OK, Server nginx/1.30.4, Content-Type text/html and Content-Length 499. The page heading was Hey me!.

Browser and local computer tests

The page opened at http://13.60.73.156/. A separate curl test from the local computer returned HTTP 200 and the expected HTML, verifying access through the security group.

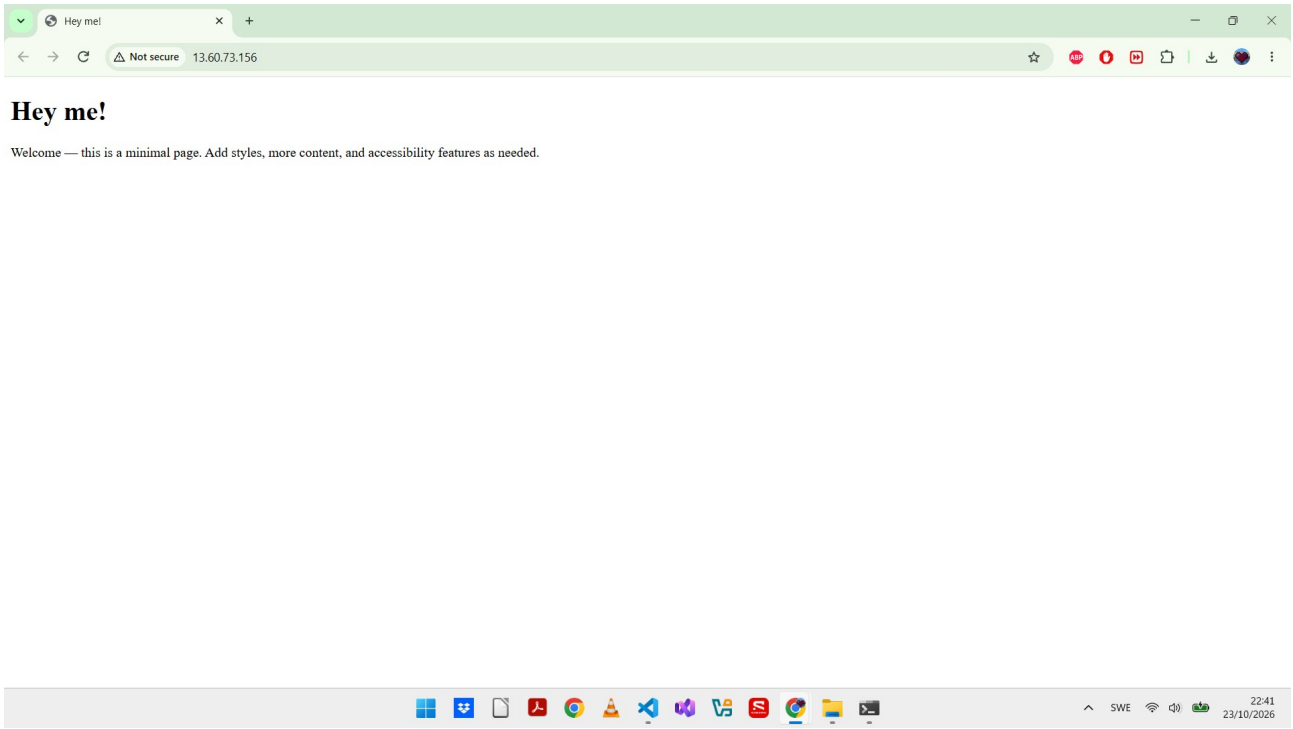


Figure 9. Website displayed at the instance public IP

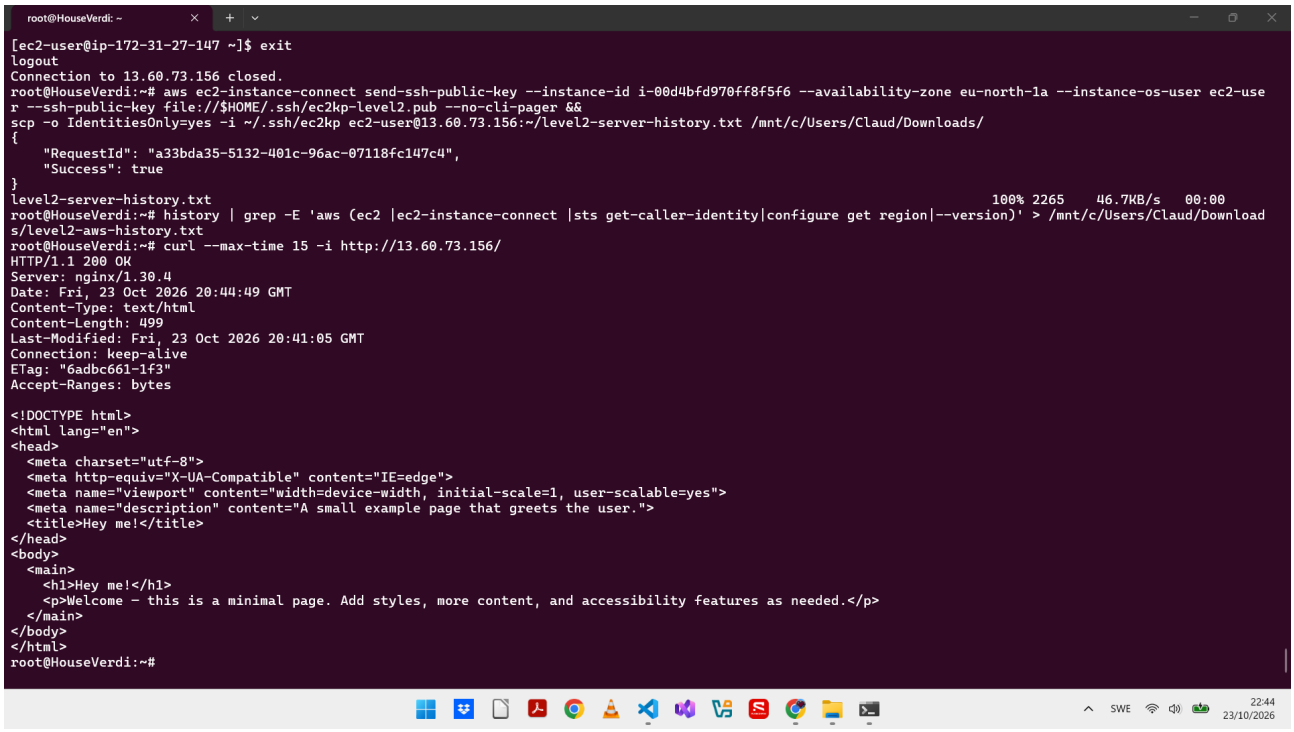


Figure 10. HTTP 200 from the local WSL terminal

DNS and automated content test

I retrieved PublicDnsName with AWS CLI and stored it in MY_EC2. The address was ec2-13-60-73-156.eu-north-1.compute.amazonaws.com. Both the HTTP test against the DNS name and the content test succeeded.

```
export MY_EC2=$(aws ec2 describe-instances \
  --instance-ids i-00d4bfd970ff8f5f6 \
  --query 'Reservations[0].Instances[0].PublicDnsName' --output text)
curl --max-time 15 -I "http://$MY_EC2/"
curl -fsS --max-time 15 "http://$MY_EC2/" | \
  grep -qi 'Hey me!' && echo "YES"
```

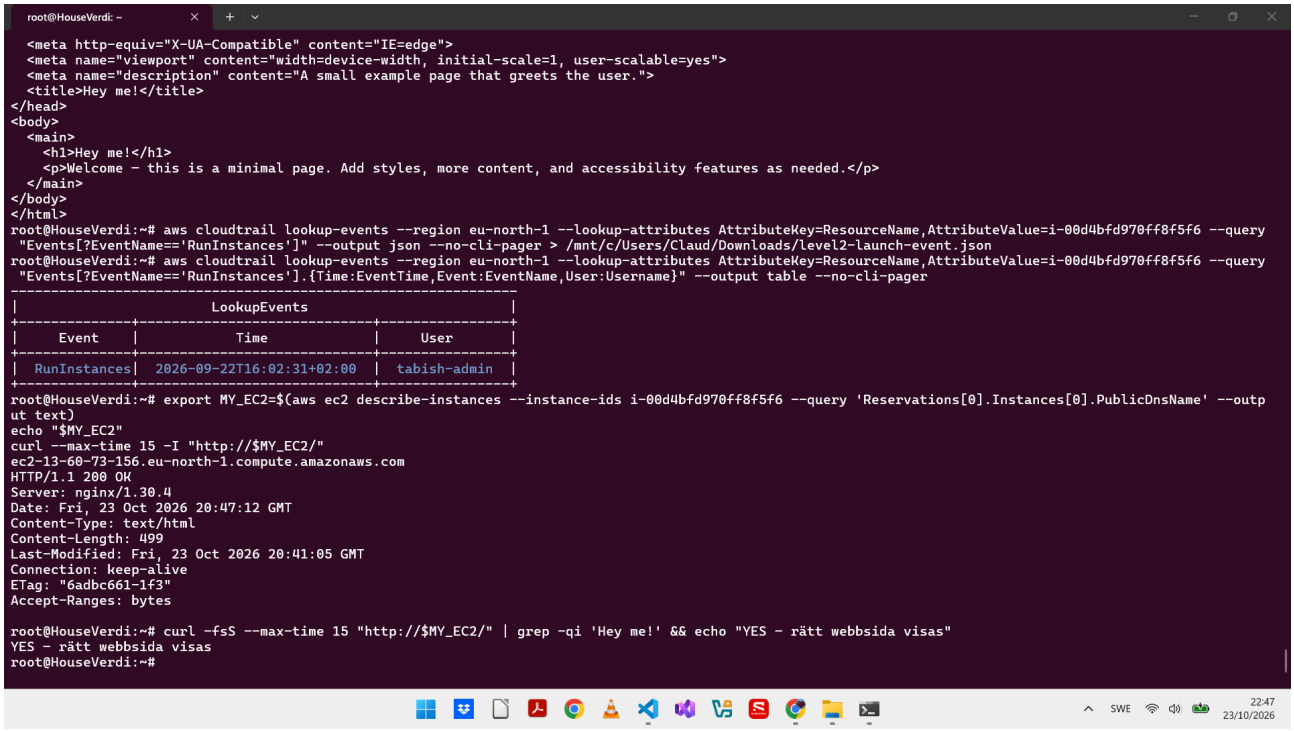


Figure 11. DNS test with HTTP 200 and successful content test

The test was adapted to the actual page text using a case-insensitive search. The terminal printed YES followed by the Swedish text meaning the correct page is displayed.

Termination and verified cleanup

After saving the documentation and tests, I checked the instance ID, name and state. I then terminated level2-web-server using AWS CLI. The response showed the transition from running to shutting-down.

A separate state check returned terminated on 23 October 2026, completing the guide's final step. The website at the previous IP address is no longer running on this instance.

```
aws ec2 terminate-instances --region eu-north-1 \
  --instance-ids i-00d4bfd970ff8f5f6 --no-cli-pager
aws ec2 describe-instances --region eu-north-1 \
  --instance-ids i-00d4bfd970ff8f5f6 \
  --query 'Reservations[0].Instances[0].State.Name' \
  --output text --no-cli-pager
```

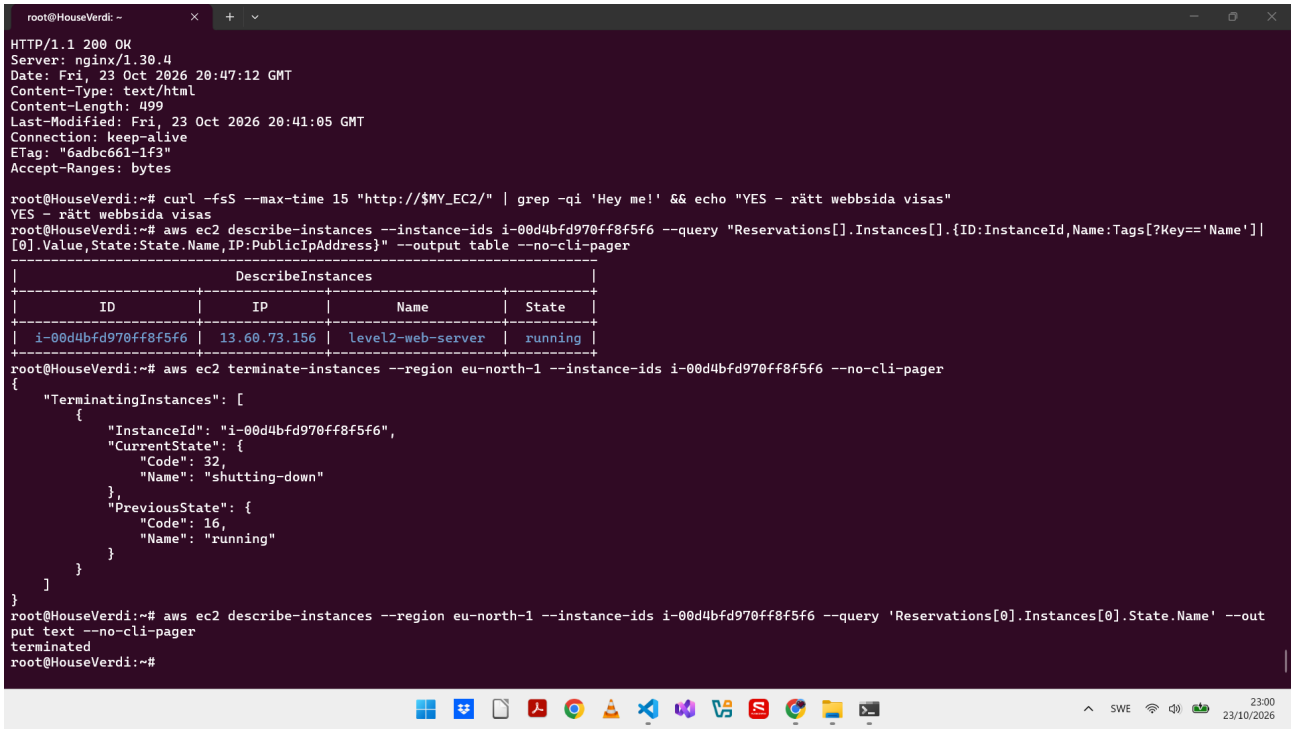


Figure 12. AWS CLI termination and verified final state terminated

Problems solutions and conclusion

Problems and solutions

The missing SSH key was handled through EC2 Instance Connect. The missing Nginx service was resolved by installation and startup. The guide’s file copy was adapted because the repository contained a file named heyme, which needed to be deployed as index.html.

Result

This completion demonstrates verified AWS CLI access, network checks, SSH login, installation, manual deployment and successful web tests. CloudTrail supplements the terminal history with evidence of the original CLI instance creation.

Improvements and cleanup

Nginx was enabled for automatic startup. The security group already restricted HTTP and SSH to the current client address. HTTPS, automated deployment and a reboot test are possible next improvements.

The guide’s final step was completed by terminating the instance and verifying the terminated state. Screenshots preserve the web test results and cleanup evidence.

Reference and evidence

level2.console.complete (4)(2).sh is the workflow reference.

level2-aws-history.txt contains local AWS commands. Older commands before the current Level 2 sequence relate to other attempts.

level2-server-history.txt contains server commands. The initial Level 3 entries are older history and are not used as evidence of the Level 2 result.

level2-launch-event.json contains the CloudTrail RunInstances event.

Embedded screenshots were captured on 23 October 2026 and show terminal results and the deployed page.